ANALYTICS PROJECT REPORT

E-Commerce Customer & Marketing Analytics

End-to-End Analysis using SQL, Python, Excel & Power BI

Dataset: Sample Superstore Dataset — Kaggle

PYTHON SQL EXCEL POWER BI

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WHY THIS PROJECT

Objective

Build a complete analytics pipeline — from raw data to executive dashboards — to uncover the drivers of sales and profitability and turn them into concrete business actions.

1

Clean & Prepare Data

Handle missing values, duplicates, data types and outliers in Python.

2

Explore the Data

Run EDA to understand distributions and relationships between Sales, Profit, Discount & Quantity.

3

Build an Excel Dashboard

Deliver a fast, spreadsheet-native KPI view for business users.

4

Query with SQL

Answer targeted business questions with precise, ad-hoc aggregation.

5

Design a Power BI Report

Create an interactive, filterable, executive-level dashboard.

6

Synthesize Insights

Turn findings into prioritized, actionable recommendations.

THE CHALLENGE

Business Problem

Superstore sells across Furniture, Office Supplies and Technology in four U.S. regions. Order volume and sales have grown — but profitability hasn't kept pace.

Management lacks a unified, data-driven view connecting sales, discounting behaviour and profitability — so decisions on discounting, regional investment and category focus are made without knowing which combinations actually erode margin.

This report identifies where profit is made and lost, quantifies the role of discounting, and translates the findings into prioritized action.

ORDERS SOLD AT A LOSS

18.73%

nearly 1 in 5 orders

AVG. DISCOUNT	PROFIT MARGIN
15.63%	**~12.3%**

Discount correlates negatively with Profit (r = -0.22) — the strongest controllable driver of loss.

THE DATA

Dataset Overview

Sample Superstore Dataset (Kaggle) — multi-year U.S. retail order-line transactions.

9,977	37,820	₹22.96L	₹2.86L
Orders	Units Sold	Total Sales	Total Profit

STRUCTURE OF THE DATA

- **Categories:** Furniture · Office Supplies · Technology
- **Regions:** Central · East · South · West
- **Segments:** Consumer · Corporate · Home Office
- **Ship Modes:** Standard · Second · First Class · Same Day

KEY FIELDS

Sales / Profit	Numeric — revenue & margin
Discount	Numeric — 0% to 80%
Quantity	Numeric — units per line
Order / Ship Date	Date fields
Category / Sub-Category	Product classification

HOW IT WAS BUILT

Project Workflow

A four-stage pipeline turns raw transactions into decisions.

1 Python
Clean data & run EDA

→

2 SQL
Query business questions

→

3 Excel
Build KPI dashboard

→

4 Power BI
Interactive exec. dashboard

→

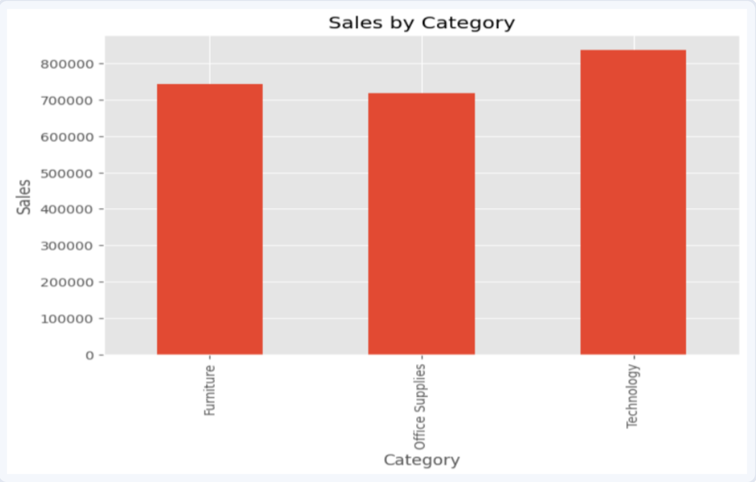
5 Insights
Recommend & act

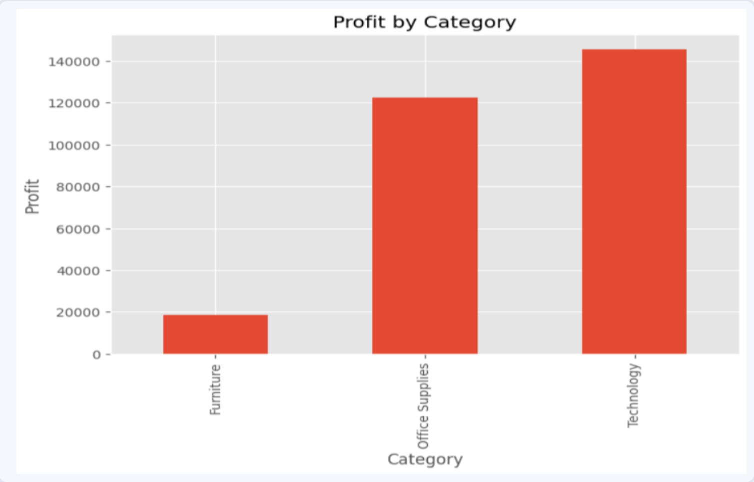
Each stage cross-validates the last: Python's EDA findings reappear in SQL aggregates, and both are visualized identically across the Excel and Power BI dashboards — giving high confidence in the final recommendations.

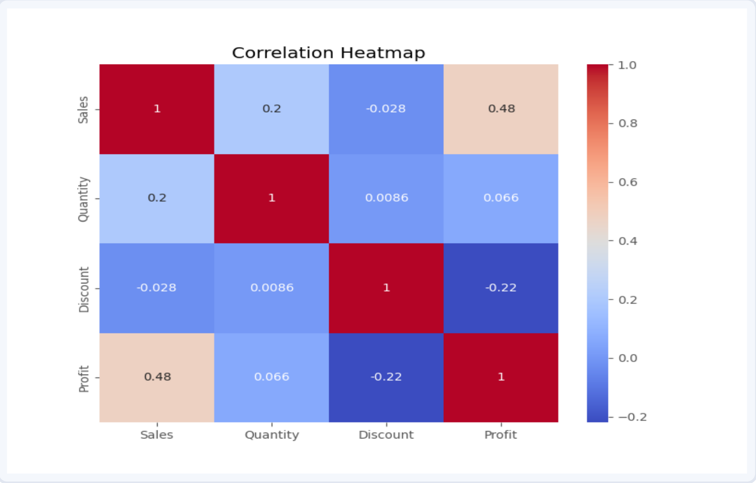
Python Cleaning & EDA

Cleaning & Key Findings

- ✓ Removed duplicates, fixed dtypes, converted dates
- ✓ Validated Discount range & handled Profit outliers
- ✓ Sales & Profit are right-skewed — small orders dominate volume
- ✓ Discount ↔ Profit correlation: -0.22 (negative)
- ✓ Sales ↔ Profit correlation: +0.48 (positive)



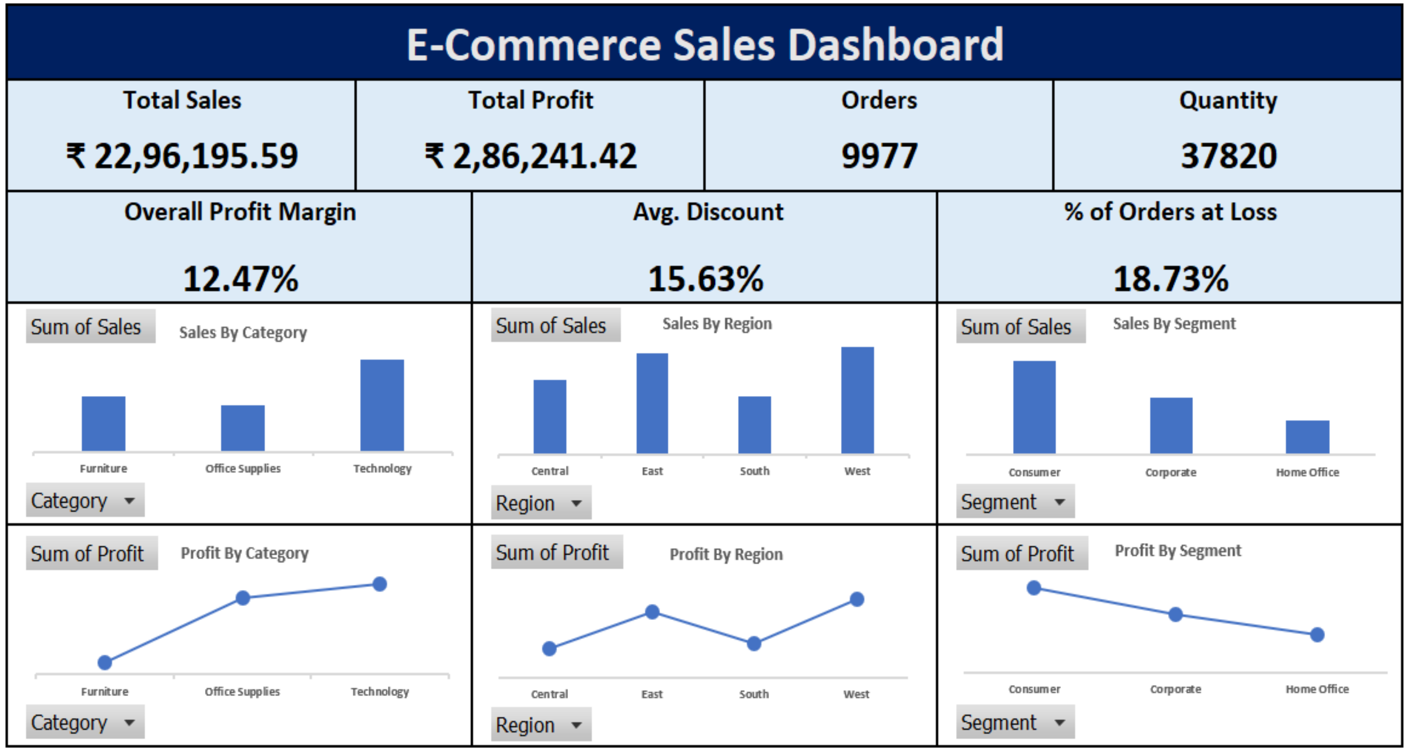




SPREADSHEET REPORTING

Excel Dashboard

PivotTable / PivotChart based KPI dashboard, filterable by Category, Region & Segment.



Total Sales

₹22,96,195.59

Total Profit

₹2,86,241.42

Overall Margin

12.47%

Avg. Discount

15.63%

% Orders at Loss

18.73%

QUERYING THE DATA

SQL Analysis

Precise, ad-hoc aggregation used to answer targeted business questions.

Loss-Making Sub-Categories

```
SELECT Sub_Category,
       SUM(Sales)  AS Total_Sales,
       SUM(Profit) AS Total_Profit
FROM superstore
GROUP BY Sub_Category
HAVING SUM(Profit) < 0
ORDER BY Total_Profit ASC
LIMIT 10;
```

Finding: Binders (-31K), Tables (-21K) and Machines (-19K) are the largest loss contributors — despite strong sales volume.

QUERIES ANSWERED

Category profit & margin

Technology leads on profit & margin

Top loss sub-categories

Binders, Tables, Machines

Regional performance

West & East outperform

Discount impact

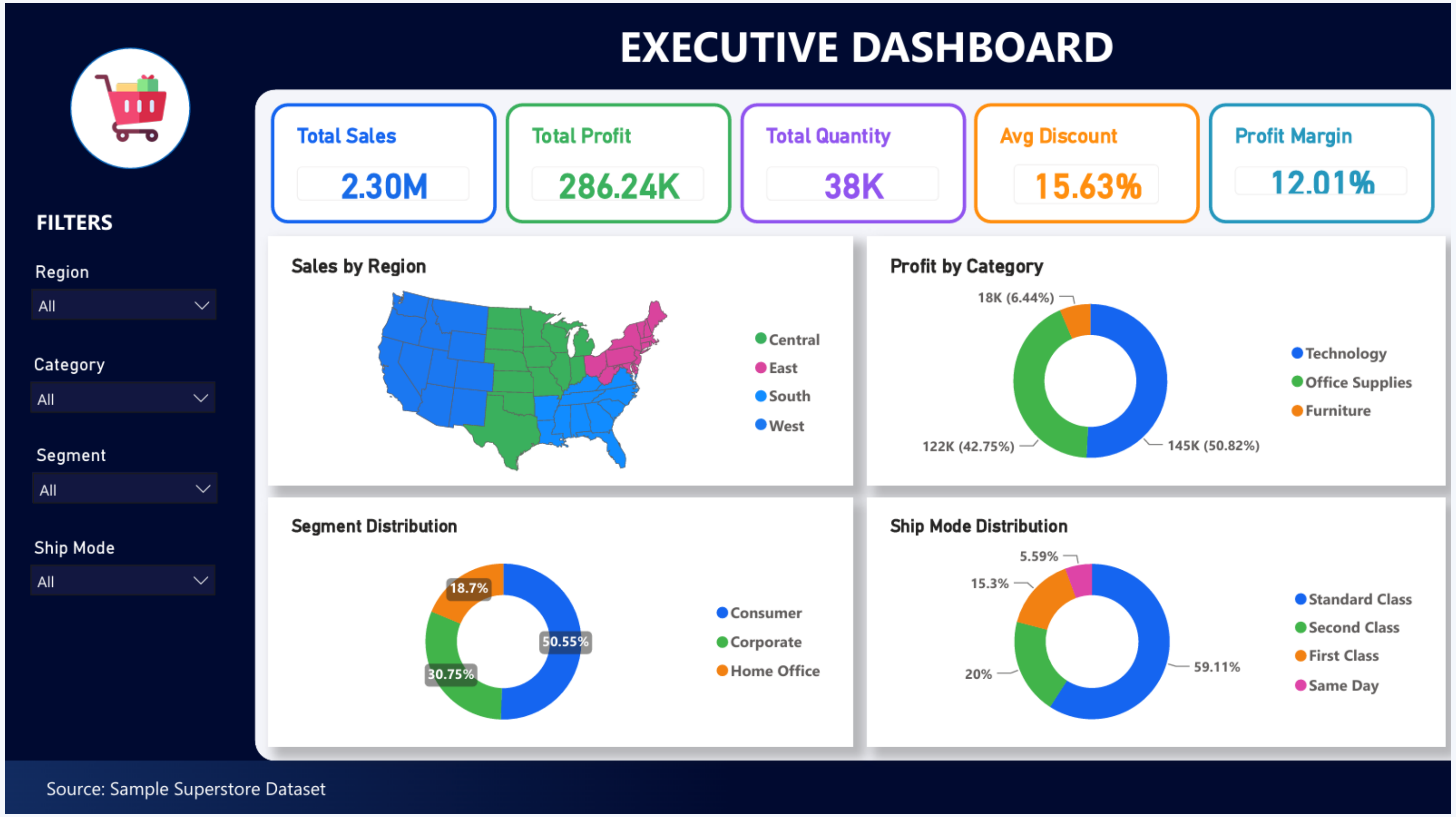
Avg. profit turns negative past ~20–30% discount

Loss-order rate

18.73% of orders unprofitable

Power BI Dashboard

Executive Dashboard



Top-Line KPIs

Total Sales 2.30M, Total Profit 286.24K, 38K units, Avg. Discount 15.63%, Profit Margin 12.01%.

Regional Sales Map

Filled U.S. map showing sales concentration by Region.

Category & Segment Mix

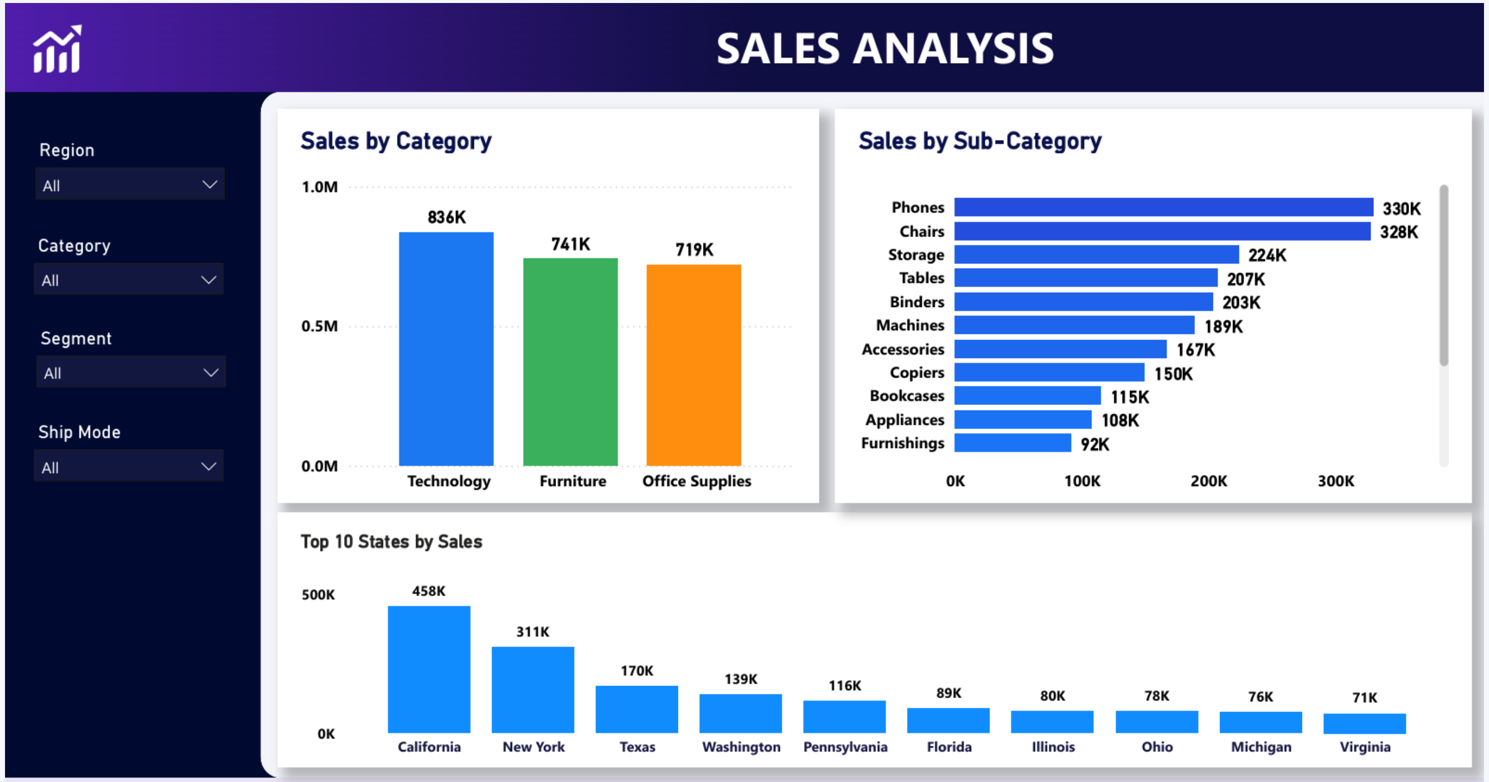
Profit by Category donut: Technology 50.8%, Office Supplies 42.8%, Furniture 6.4%; Segment split led by Consumer at 50.6%.

Ship Mode Distribution

Standard Class dominates fulfillment at 59.1% of shipments.

Power BI Dashboard

Sales Analysis



Source: Sample Superstore Dataset

Sales by Category

- Technology leads (836K), ahead of Furniture (741K) and Office Supplies (719K).

Sales by Sub-Category

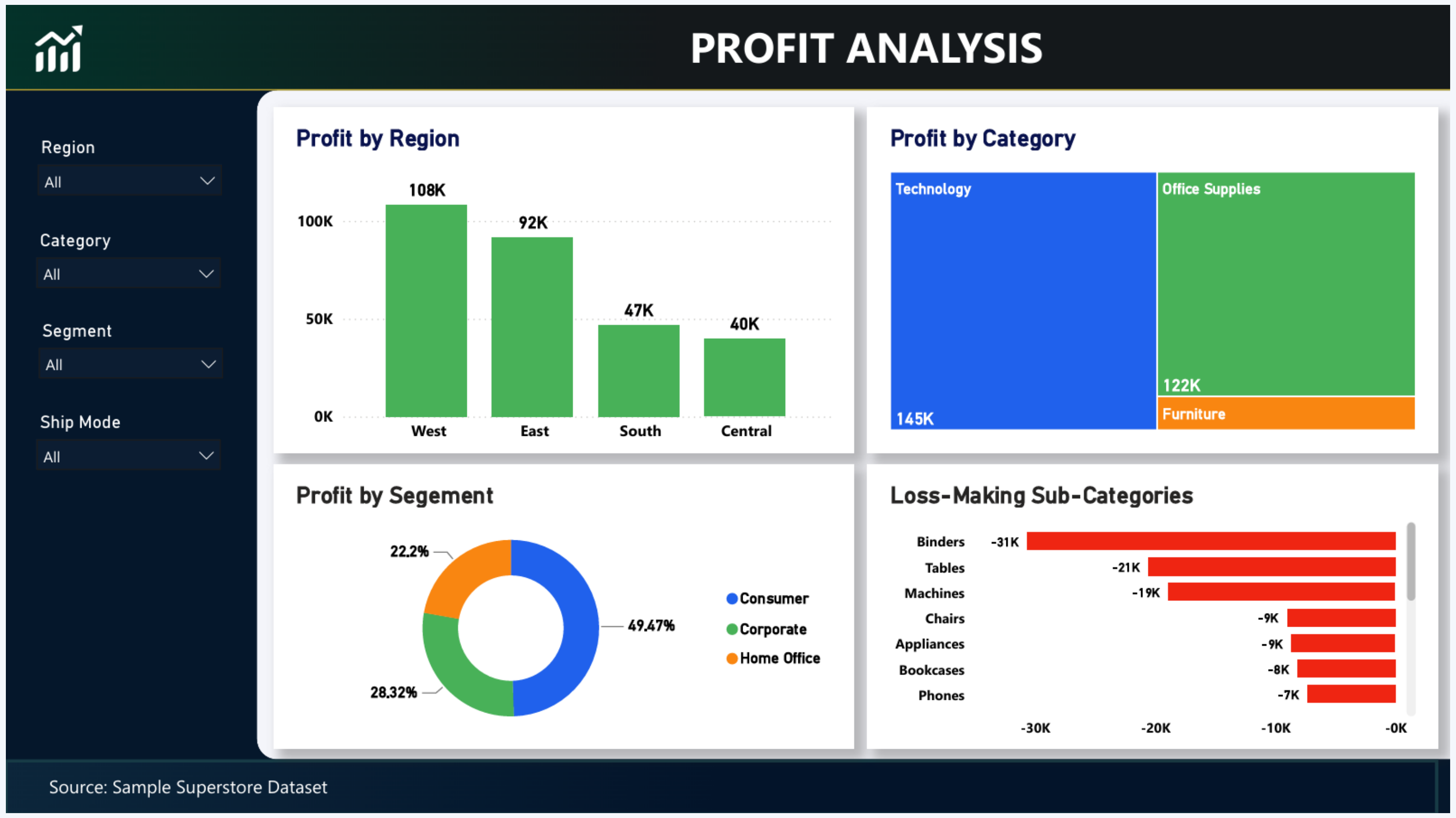
- Phones (330K) and Chairs (328K) are the top two sub-categories, down to Furnishings (92K).

Top 10 States

- California (458K) and New York (311K) are by far the largest state markets, ahead of Texas, Washington & Pennsylvania.

Power BI Dashboard

Profit Analysis



Profit by Region

West (108K) and East (92K) rank well ahead of South (47K) and Central (40K).

Profit by Category

Technology (145K) and Office Supplies (122K) dominate; Furniture contributes only a sliver.

Profit by Segment

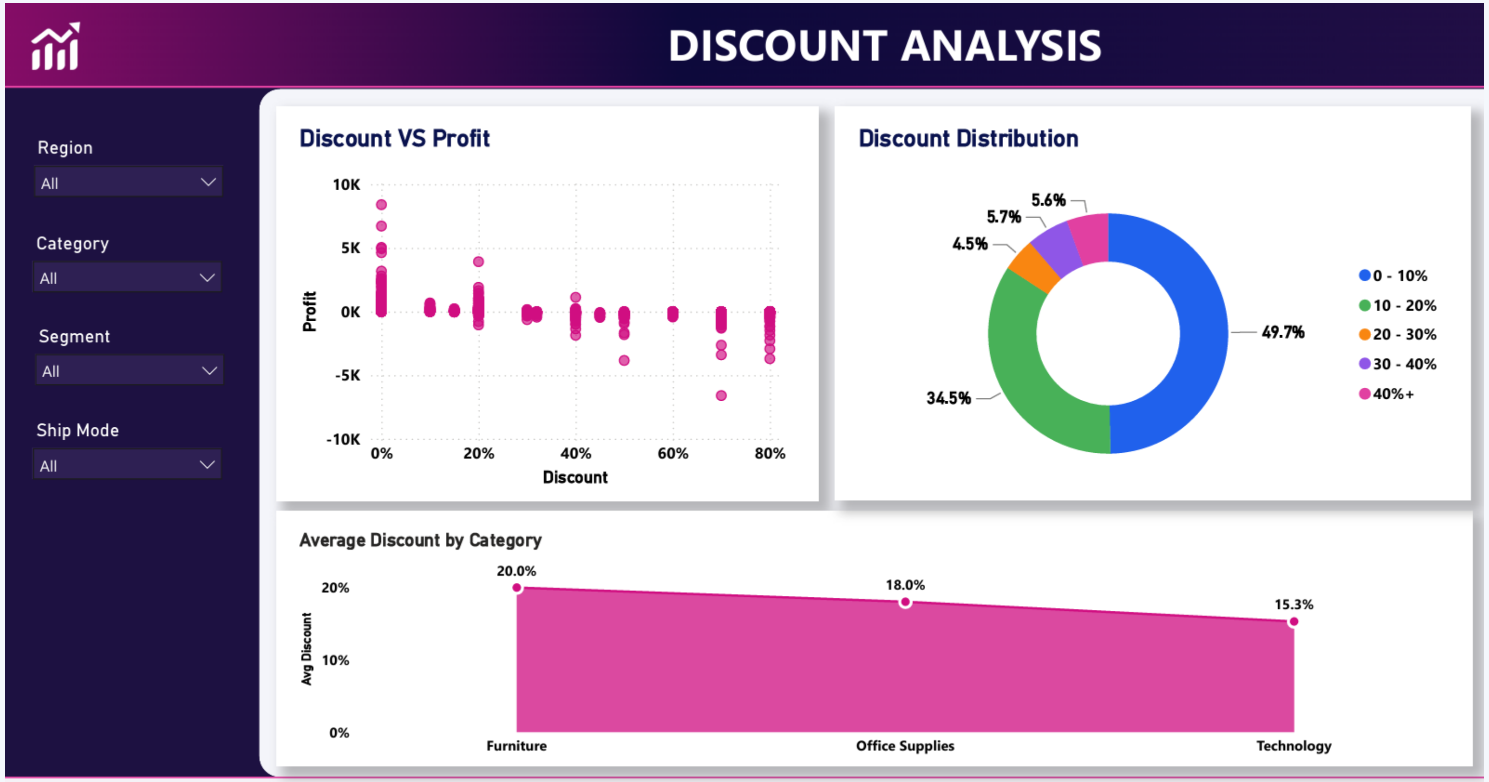
Consumer 49.5%, Corporate 28.3%, Home Office 22.2%.

Loss-Making Sub-Categories

Binders (-31K), Tables (-21K) and Machines (-19K) are the largest loss contributors.

Power BI Dashboard

Discount Analysis



Source: Sample Superstore Dataset

Discount vs Profit

Profit clusters positive near 0% discount and trends toward loss as discount rises, especially past 60-70%.

Discount Distribution

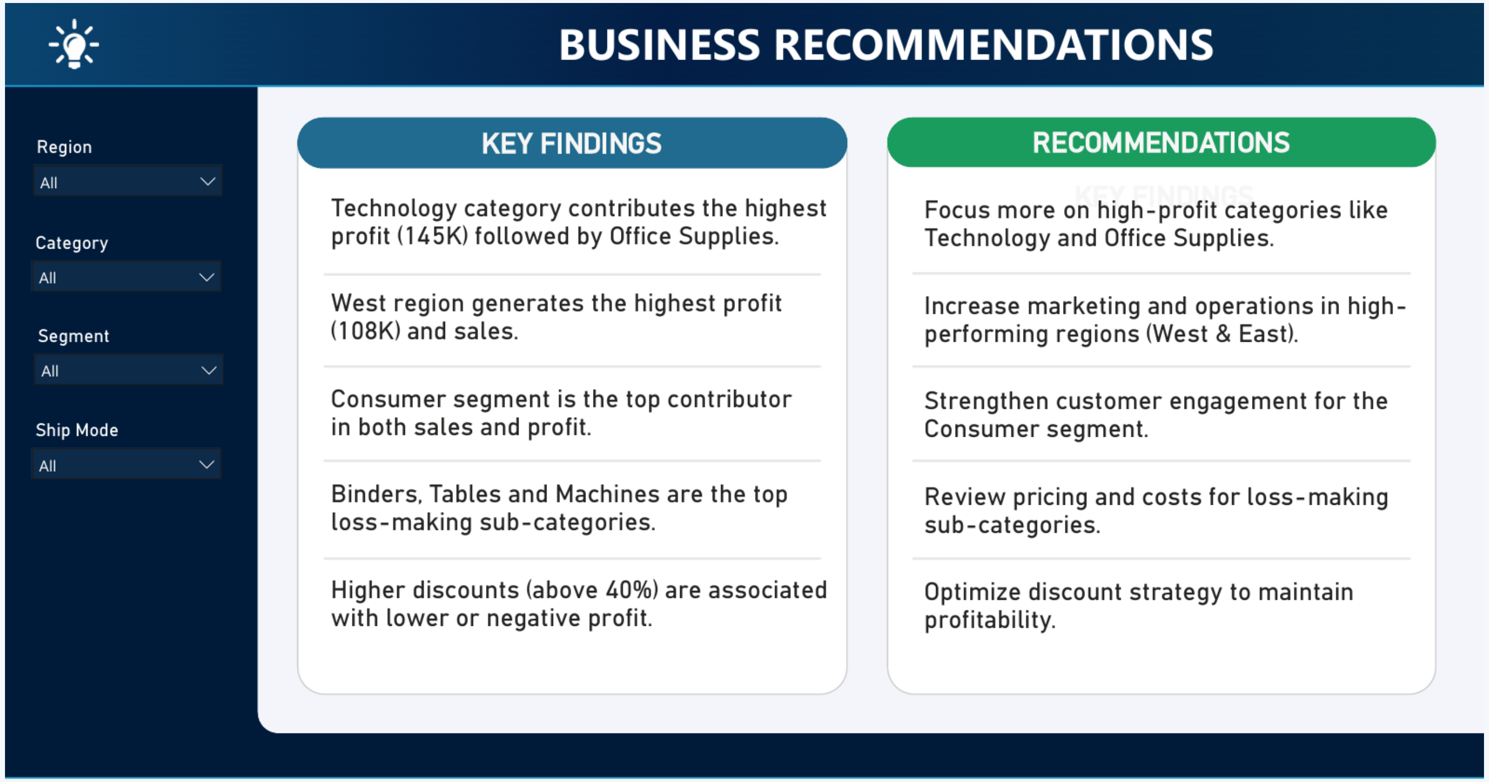
49.7% of orders carry 0-10% discount; 34.5% carry 10-20%; a combined 15.8% receive 20%+.

Avg. Discount by Category

Furniture receives the steepest average discount (20.0%), ahead of Office Supplies (18.0%) and Technology (15.3%).

Power BI Dashboard

Business Recommendations Page



Source: Sample Superstore Dataset

Findings ↔ Actions

- Two-column layout pairs each key finding directly with its recommended action.

Built for Stakeholders

- Lets business users see not just what the data shows, but what it implies.

Closes the Loop

- Mirrors the structure this presentation itself follows — insights, then recommendations.

WHAT THE DATA TELLS US

Key Insights

Technology drives profit

50.8% of total profit; lowest average discount of the 3 categories.

Furniture underperforms

Nearly matches Technology on sales but earns only ~6% of profit.

West & East lead regions

Together contribute 60%+ of sales and ~200K of total profit.

Consumer segment dominates

49.5% of total profit — nearly half — vs. Corporate & Home Office.

Discounting erodes margin

Correlation of -0.22; profit turns negative past ~20–30% discount.

18.73% of orders lose money

A quantifiable, addressable drag on overall profitability.

3 sub-categories bleed profit

Binders, Tables & Machines are chronic loss-makers.

Standard Class dominates

59.1% of shipments — concentrated logistics cost structure.

WHAT TO DO NEXT

Recommendations

1. **Tighten discount governance**

   Cap/require approval above 20–30% discount.

2. **Review loss-making sub-categories**

   Re-price Binders, Tables, Machines & others.

3. **Re-balance Furniture discounting**

   Reduce blanket discounts; use bundling instead.

4. **Double down on Technology & Office Supplies**

   Weight marketing & inventory investment here.

5. **Sustain West & East investment**

   Investigate Central & South underperformance first.

6. **Deepen Consumer engagement**

   Loyalty & retention programs for the top segment.

7. **Track the loss-order rate**

   Make 18.73% a recurring KPI on both dashboards.

WHY IT MATTERS

Business Impact

Implementing the recommendations targets the two biggest controllable levers on profit: discount policy and loss-making sub-categories.

REDUCE

Loss-Order Rate

from 18.73% via discount caps & pricing review

PROTECT

Margin on Furniture

by cutting its 20% avg. discount toward category norms

RECOVER

Sub-Category Losses

from Binders, Tables & Machines through re-pricing

GROW

High-Margin Categories

by shifting investment toward Technology & Office Supplies

BRINGING IT TOGETHER

Conclusion

Superstore's revenue base is healthy and diversified — but profitability is being materially eroded by uncontrolled discounting and a small set of chronically loss-making sub-categories, particularly within Furniture.

Technology, Office Supplies, the West & East regions, and the Consumer segment are Superstore's core strengths and should anchor future growth — while tightening discount policy and re-pricing loss-makers represents a concrete, sizable opportunity to lift margin without needing any increase in sales volume.

Four analytical layers — Python, SQL, Excel and Power BI — cross-validated one another at every step, giving high confidence in these findings.

WHAT'S NEXT

Future Enhancements

Sales & profit forecasting

Time-series models (Prophet/ARIMA) by category & region.

Customer segmentation

RFM analysis / clustering for high-value customer targeting.

Discount optimization

Price-elasticity model for sub-category-level discount caps.

Automated ETL pipeline

Scheduled refresh from Python → SQL → Power BI.

Drill-through reporting

Region/category drill-down to product & customer level.

External data enrichment

Marketing spend, competitor pricing & macro indicators.

THANK YOU

Questions & Discussion

E-Commerce Customer & Marketing Analytics — SQL · Python · Excel · Power BI

PYTHON

SQL

EXCEL

POWER BI